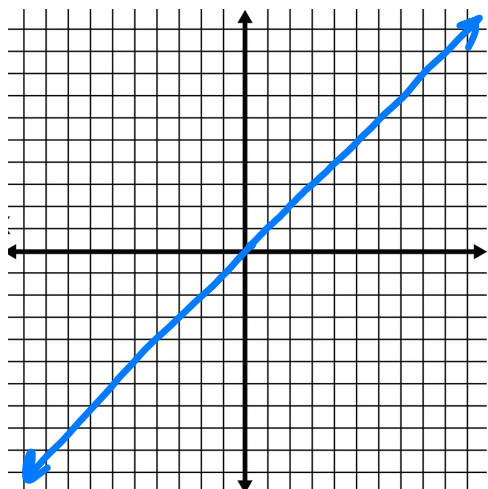


key

Graph Library



Today we will look at six parent functions and explore their basic properties. Draw each of the graphs below without the aid of your calculator. Fill in at least three ordered pairs in your table and determine the domain and range of each curve. Find both intercepts, as well as the intervals on which the function is increasing and decreasing.



$$f(x) = x$$

identity function

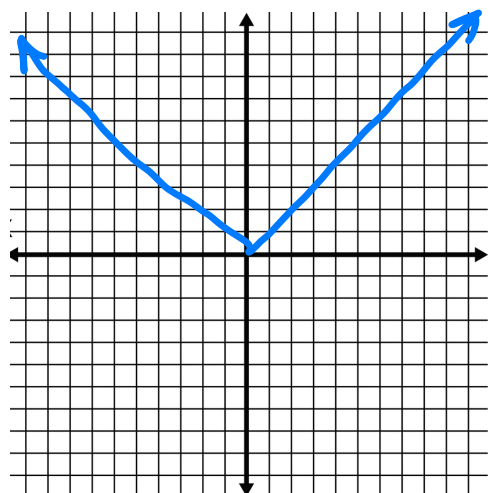
x	$f(x)$
-3	-3
1	1
7	7
10	10

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

X-intercept: 0 Y-intercept: 0

Increasing on $(-\infty, \infty)$ Decreasing on never



$$f(x) = |x|$$

absolute value function

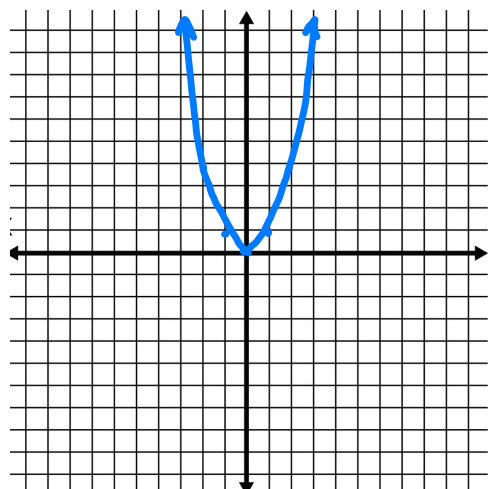
x	$f(x)$
-3	3
-1	1
2	2
6	6

Domain: $\mathbb{R}$

Range: $y \geq 0$

X-intercept: 0 Y-intercept: 0

Increasing on $(0, \infty)$ Decreasing on $(-\infty, 0)$



$$f(x) = x^2$$

quadratic function

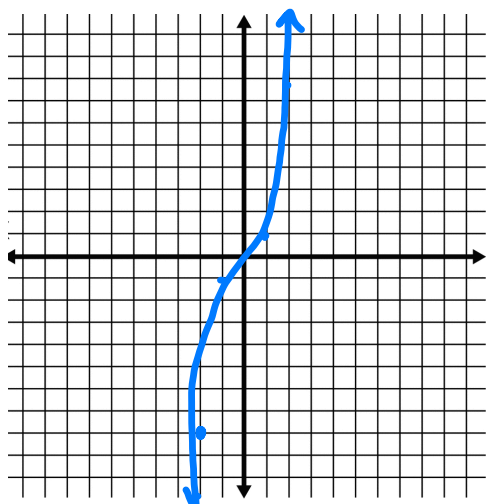
x	$f(x)$
-1	1
0	0
1	1
2	4

Domain: $\mathbb{R}$

Range: $y \geq 0$

X-intercept: 0 Y-intercept: 0

Increasing on $(0, \infty)$ Decreasing on $(-\infty, 0)$



$$f(x) = x^3$$

cubic
function

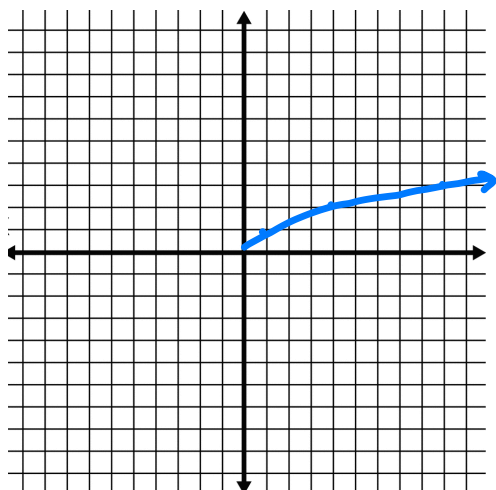
x	$f(x)$
0	0
1	1
2	8
3	27

Domain: $\mathbb{R}$

Range: $\mathbb{R}$

X-intercept: 0 Y-intercept: 0

Increasing on $(-\infty, \infty)$ Decreasing on never



$$f(x) = \sqrt{x}$$

root function
or radical

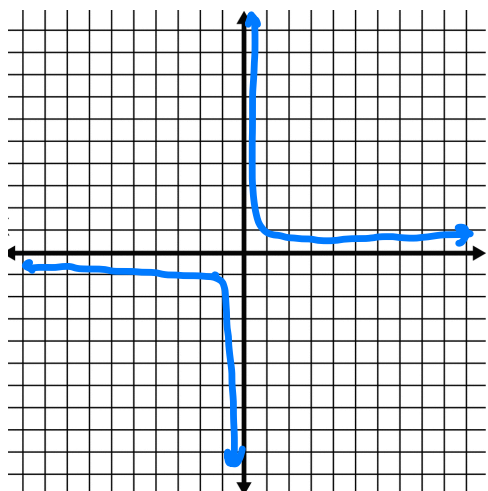
x	$f(x)$
0	0
1	1
4	2
9	3

Domain: $x \geq 0$

Range: $y \geq 0$

X-intercept: 0 Y-intercept: 0

Increasing on $(0, \infty)$ Decreasing on never



$$f(x) = \frac{1}{x}$$

reciprocal
function

x	$f(x)$
1	1
-3	$-\frac{1}{3}$
2	$\frac{1}{2}$
4	$\frac{1}{4}$

Domain: $\{x \in \mathbb{R} \mid x \neq 0\}$

Range: $\{y \in \mathbb{R} \mid y \neq 0\}$

X-intercept: none Y-intercept: none

Increasing on never Decreasing on $(-\infty, 0) \cup (0, \infty)$

Lesson 1.4 – Library of Parent Functions

QuickNotes

A parent function is the simplest function of a family of functions.

Families of functions (quadratic, linear, radical, etc.) share similar algebraic properties, graphs, and behaviors.

Check Your Understanding

1. Which parent function(s) demonstrate a constant rate of change?

Identity function

Absolute value

2. Identify the parent function in each of the following equations:

a. $f(x) = \frac{2}{x-3}$ $1/x$

b. $g(x) = 4x^2 + 5x^3 - 1$ x^3

c. $h(x) = -|7x - 4|$ $|x|$

3. Determine if the relationship is linear, quadratic, or neither. How do you know?

a.

x	y
-7	20
-5	18
-3	14
-1	8
1	0

$+2 (-7, 20) \rightarrow (-2) - 2$
 $+2 (-5, 18) \rightarrow (-4) - 2$
 $+2 (-3, 14) \rightarrow (-6) - 2$
 $+2 (-1, 8) \rightarrow (-8) - 2$
 $+2 (1, 0) \rightarrow (-8) - 2$

Quadratic because there is a common second difference.

b.

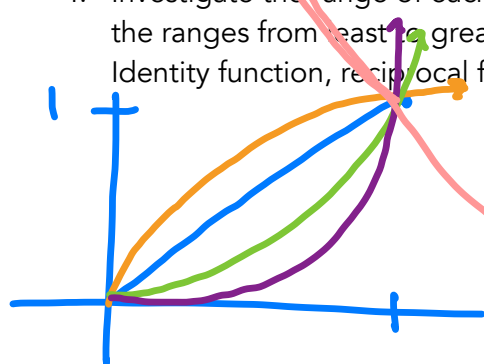
Hours studied	Points earned on test
0.5	62
1	70
2	86
2.5	92
3	98

$0.5 (-7, 20) \rightarrow 8$
 $1 (-5, 18) \rightarrow 16$
 $2 (-3, 14) \rightarrow 6$

Neither, the linear pattern of 16 pts per hour of study is not consistent throughout.

4. Investigate the range of each parent function below over the interval $0 < x < 1$. Then order the ranges from least to greatest.

Identity function, reciprocal function, quadratic function, cubic function, square root function



— cubic
 — quadratic
 — identity
 — square root

— reciprocal

cubic, quad, identity, square root, reciprocal